Unit 2 Guided Notes — Exponents & Scientific Notation

ANSWER KEY

Same base? Move the exponents. Big or tiny number? Pack it into a × 10^n.

🎯 **By the end of this unit, I can…**

- Apply the product, quotient, and power rules of exponents
- Simplify expressions with negative and zero exponents
- Write and compute with numbers in scientific notation

Standards: NC.8.EE.1 · NC.8.EE.3 · NC.8.EE.4

Answer key: every blank and final answer is filled in green.

1 Exponent Rules

📖 **Words to know**

Base: the number being multiplied. In 2^5, the base is 2.

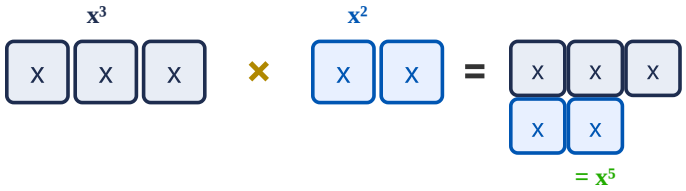
Exponent: the small number up high. It tells how many times to multiply the base. In 2^5, the exponent is 5.

Coefficient: the number in front of a variable. In $4x^3$, the coefficient is 4.

Monomial: one term: a number, a variable, or a product of them, like $4x^3$ or $-2xy^2$.

📌 **KEY RULE:** Exponent rules only apply when the BASES are the SAME!

Product Rule — Count the Factors



$x^3 \times x^2 = x^{3+2} = x^5$

Count the factors: 3 + 2 = 5 → ADD the exponents

The Five Core Rules

Rule	Formula	Simple Example	Monomial Example	Key Phrase
Product Rule	$x^a \times x^b = x^{a+b}$	$x^3 \cdot x^4 = x^7$	$(2x^3)(3x^2) = 6x^5$	Same base? ADD
Quotient Rule	$x^a \div x^b = x^{a-b}$	$x^6 \div x^2 = x^4$	$(12x^7) \div (3x^2) = 4x^5$	Same base? SUBTRACT
Power Rule	$(x^a)^b = x^{a \cdot b}$ (multiply the exponents)	$(x^2)^3 = x^6$	$(2x^3)^2 = 4x^6$	Power to power? MULTIPLY
Zero Exponent	$x^0 = 1$	$5^0 = 1$	$(3x^5)^0 = 1$	ANY base⁰=1
Negative Exponent	$x^{-n} = \frac{1}{x^n}$	$2^{-3} = 1/8$	$4x^{-3} = 4/x^3$	Flip to fraction

⚠️ **PARENTHESES MATTER!** -3^2 = **−9** vs. $(-3)^2$ = **9**. Without ( ), ONLY the base gets the exponent — not the negative sign!

Modeling the Rules — See WHY

Each rule comes from writing out the repeated multiplication.

Product Rule · $x^3 \times x^2 = x^5$

$x \cdot x \cdot x \times x \cdot x = x \cdot x \cdot x \cdot x \cdot x = x^5$. Count the factors: 3 + 2 = 5. Same base → ADD the exponents.

Quotient Rule · $x^5 \div x^2 = x^3$

$x \cdot x \cdot x \cdot x \cdot x \div x \cdot x$ → cancel 2 matching pairs → $x \cdot x \cdot x = x^3$. Cancel matching factors: 5 − 2 = 3. Same base → SUBTRACT the exponents.

Power Rule · $(x^2)^3 = x^6$

Three copies of x^2: $(x \cdot x)(x \cdot x)(x \cdot x) = x^6$. 2 × 3 = 6. Power to a power → MULTIPLY.

Zero Exponent · $x^0 = 1$

Count down by 2s: $2^3 = 8$, $2^2 = 4$, $2^1 = 2$. Each time the exponent drops by 1, divide by 2. So $2^0 = 2 \div 2 = 1$. This works for any base, so $x^0 = 1$.

Negative Exponent · $x^{-3} = \frac{1}{x^3}$

$x^2 \div x^5$: cancel 2 pairs of x. 3 x's are left on the bottom. So $x^{-3} = \frac{1}{x^3}$. A negative exponent means flip it to the bottom: $x^{-n} = \frac{1}{x^n}$.

Quick Check

MC Simplify $x^5 \cdot x^3$. Which rule applies, and what is the result?

- ● **x^8 (Product Rule)**
- ○ x^{15} (Power Rule)
- ○ x^2 (Quotient Rule)

Match the Rule to its Key Phrase

MATCH Write the letter of each rule's key phrase on the line.

B	1. Product Rule	**A.**	Power to power? MULTIPLY
D	2. Quotient Rule	**B.**	Same base? ADD exponents
A	3. Power Rule	**C.**	Flip to a fraction
C	4. Negative Exponent	**D.**	Same base? SUBTRACT exponents

Worked Example — Simplify $(4x^3y^2)(3x^5y)$

Step 1 — Multiply coefficients: $4 \times 3 = 12$

Step 2 — Add x exponents: $x^3 \cdot x^5 = x^8$

Step 3 — Add y exponents: $y^2 \cdot y^1 = y^3$

✓ **Step 4 — Combine:** $12x^8y^3$

ERROR ANALYSIS Simplify $(3x^2)^3$. A student wrote $3x^6$. What went wrong?

The coefficient 3 must ALSO be cubed: $3^3 = 27$. Correct: $27x^6$. The power rule applies to EVERYTHING inside the parentheses.

2 Scientific Notation

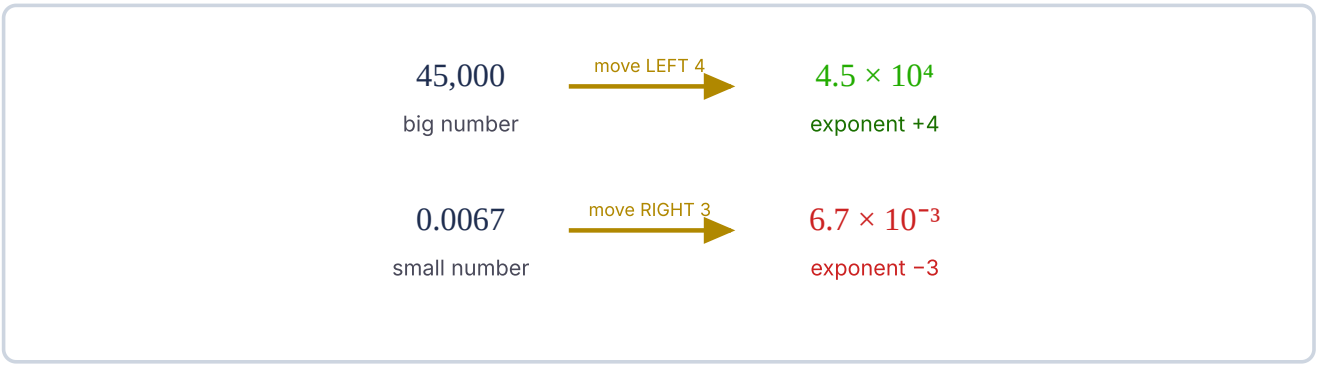
📖 **Words to know**

Standard form: a number written out with all its digits, like 45,000,000.

Scientific notation: a number written as (a number from 1 up to 10) × 10 to a power, like 4.5×10^7.

📌 **THE FORMAT:** A number in scientific notation looks like 4.5×10^7. The first number must be at least 1 and less than 10. The base is always 10.

Move the Decimal



Type	Direction	Exponent	Example
Large number	Move decimal LEFT	**POSITIVE**	45,000 = 4.5 × 10^4
Small number (< 1)	Move decimal RIGHT	**NEGATIVE**	0.0067 = 6.7 × 10^{-3}

Multiplying — 3-Step Process

1. **MULTIPLY** the coefficients (the a-values)
2. **ADD** the exponents
3. Adjust if coefficient is NOT between **1** and **10**

Example (no adjustment): (3 × 10^4) × (2 × 10^5)

Multiply coefficients: 3 × 2 = 6

ADD exponents: 4 + 5 = 9

✓ **Answer: 6 × 10^9**

Dividing — 3-Step Process

1. **DIVIDE** the coefficients
2. **SUBTRACT** the exponents
3. Adjust if needed

Example (no adjustment): (8 × 10^6) ÷ (4 × 10^2)

Divide coefficients: 8 ÷ 4 = 2

SUBTRACT exponents: 6 − 2 = 4

✓ **Answer: 2×10^{4}**

⚠️ **ADJUSTMENT RULES:** If coefficient ≥ 10: move decimal LEFT, ADD 1 to exponent. If coefficient < 1: move decimal RIGHT, SUBTRACT 1 from exponent.

Example needing adjustment: $(5 \times 10^{3}) \times (4 \times 10^{2})$

Multiply: 5 × 4 = 20

Add exponents: 3 + 2 = 5 → result: 20×10^{5}

Adjust (20 ≥ 10): $20 = 2.0 \times 10^{1}$, so ADD 1 to the exponent

✓ **Final answer: 2.0×10^{6}**

ERROR ANALYSIS **Convert 0.0045 to scientific notation. A student wrote 4.5×10^{3}. What's the mistake?**

Small number → NEGATIVE exponent. Correct: 4.5×10^{-3}. Numbers less than 1 always have negative exponents in scientific notation.

✏ Part 2 — Practice Problems

📌 **Show every step on its own line.** Box your final answer.

A. Exponent Rules with Monomials

1) Simplify $x^3 \cdot x^5$

Answer: x^8

2) Simplify $(y^4)^3$

Answer: y^{12}

3) Simplify $a^8 \div a^3$

Answer: a^5

4) Simplify $(2x^3)(5x^2)$

Answer: $10x^5$

5) Simplify $(3m^2)^3$

Answer: $27m^6$

6) Simplify $12y^7 \div 4y^2$

Answer: $3y^5$

7) Simplify $(-2a^4)^2$

Answer: $4a^8$

8) Simplify $(5x^2y)(3xy^3)$

Answer: $15x^3y^4$

9) Simplify $18n^6 \div 6n^6$

Answer: 3

10) Simplify $(4b^3)^2 \div 2b$

Answer: $8b^5$

11) Simplify $(-3x^2)(-4x^3)$

Answer: $12x^5$

12) Simplify $(x^5y^2)^3$

Answer: $x^{15}y^6$

B. Scientific Notation Conversions

1) 45,000,000 → scientific notation

Answer: 4.5×10^7

2) 0.00032 → scientific notation

Answer: 3.2×10^{-4}

3) 6.02×10^8 → standard form

Answer: 602,000,000

4) 9.1×10^{-5} → standard form

Answer: 0.000091

5) 780,000 → scientific notation

Answer: 7.8×10^{5}

6) 0.0000067 → scientific notation

Answer: 6.7×10^{-6}

7) 3.45×10^{4} → standard form

Answer: 34,500

8) 1.2×10^{-3} → standard form

Answer: 0.0012

9) 5,280,000,000 → scientific notation

Answer: 5.28×10^{9}

10) 2.5×10^{-2} → standard form

Answer: 0.025

C. Operations

1) $(3 \times 10^{4})(2 \times 10^{5})$

Answer: 6×10^{9}

2) $(8 \times 10^{7}) \div (4 \times 10^{3})$

Answer: 2×10^{4}

3) $(5 \times 10^3)(7 \times 10^6)$

Answer: 3.5×10^{10}

4) $(9.6 \times 10^8) \div (3.2 \times 10^5)$

Answer: 3×10^3

D. Word Problems

1. The Sun is about 9.3×10^7 miles from Earth. Light travels 1.86×10^5 miles per second. How many seconds does it take light to reach Earth?

$(9.3 \times 10^7) \div (1.86 \times 10^5) = 5 \times 10^2$ seconds (500 seconds).

2. A bacterium is about 2×10^{-6} m long. A virus is about 5×10^{-8} m long. How many times longer is the bacterium?

$(2 \times 10^{-6}) \div (5 \times 10^{-8}) = 0.4 \times 10^2 = 40$ times longer.

3. The US national debt is about 3.4×10^{13} dollars. The US population is about 3.4×10^8 people. What is the debt per person?

$(3.4 \times 10^{13}) \div (3.4 \times 10^8) = 1 \times 10^5 = \$100{,}000$ per person.

4. A grain of sand has mass 6.5×10^{-4} grams. A beach has about 8×10^{18} grains of sand. What is the total mass in grams?

$(6.5 \times 10^{-4})(8 \times 10^{18}) = 52 \times 10^{14} = 5.2 \times 10^{15}$ grams.

5. There are about 3.7×10^{13} cells in a human body. Each cell is about 1×10^{-5} m wide. If all cells were lined up, how long in meters?

$(3.7 \times 10^{13})(1 \times 10^{-5}) = 3.7 \times 10^8$ meters.

6. A computer does 4.8×10^9 calculations per second. A task needs 1.44×10^{12} calculations. How many seconds to finish?

$(1.44 \times 10^{12}) \div (4.8 \times 10^9) = 0.3 \times 10^3 = 3 \times 10^2$ seconds (300 seconds).